

Math 526, F08
Quiz 6

Name: Answer
SSN: xxx-xx-

Instructions: There is totally 2 problems. You must show all of your work to receive a credit for a correct answer. No graphical calculator is allowed in the quiz and exam.

Problem 1. (10 points) Let $A = \begin{bmatrix} 1 & 3 & 1 \\ 2 & 6 & 4 \\ -2 & -6 & 0 \end{bmatrix}$.

(a) Find the dimension and a basis for the row space of A , $C(A^T)$.

Solution: $A = \begin{bmatrix} 1 & 3 & 1 \\ 2 & 6 & 4 \\ -2 & -6 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 3 & 1 \\ 0 & 0 & 2 \\ 0 & 0 & 2 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 3 & 1 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{bmatrix} = U$

$$\dim(C(A^T)) = 2$$

$$\text{One basis} = \begin{bmatrix} 1 \\ 3 \\ 1 \end{bmatrix}, \begin{bmatrix} 2 \\ 6 \\ 4 \end{bmatrix}$$

(b) Find the dimension and a basis for the nullspace of A , $N(A)$.

Solution: $\dim(N(A)) = 1$.

Set $x_2 = 1$, solve $Ux = 0$

$$\begin{aligned} x_1 + 3x_2 + x_3 &= 0 \Rightarrow x_1 + x_3 = -3 \Rightarrow x_1 = -3 \\ 2x_3 &= 0 \Rightarrow 2x_3 = 0 \Rightarrow x_3 = 0 \end{aligned}$$

$$\Rightarrow s = \begin{bmatrix} -3 \\ 1 \\ 0 \end{bmatrix}$$

$$\text{One basis of } N(A) : \begin{bmatrix} -3 \\ 1 \\ 0 \end{bmatrix}$$

(c) Are $N(A)$ and $C(A^T)$ perpendicular to each other?

Yes.

Problem 2. (10 points) Given two vectors $\mathbf{a}_1 = (1, 1, 0)$ and $\mathbf{a}_2 = (1, 1, 1)$.

(a) Find the projection matrix $\mathbf{P}$ for the subspace in $\mathbb{R}^3$ spanned by $\mathbf{a}_1$ and $\mathbf{a}_2$.

Solution: $A = \begin{bmatrix} 1 & 1 \\ 1 & 1 \\ 0 & 1 \end{bmatrix} \quad A^T = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}$

$$A^T A = \begin{bmatrix} 2 & 2 \\ 2 & 3 \end{bmatrix} \Rightarrow (A^T A)^{-1} = \frac{1}{2} \begin{bmatrix} 3 & -2 \\ -2 & 2 \end{bmatrix}$$

$$\begin{aligned} \mathbf{P} &= A (A^T A)^{-1} A^T \\ &= \begin{bmatrix} 1 & 1 \\ 1 & 1 \\ 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 3/2 & -1 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 1/2 & 1/2 & 0 \\ 1/2 & 1/2 & 0 \\ 0 & 0 & 1 \end{bmatrix} \end{aligned}$$

(b) Let $\mathbf{b} = \begin{bmatrix} 2 \\ 3 \\ 4 \end{bmatrix}$, find the projection of $\mathbf{b}$ onto the subspace by using the result from part (a).

Solution: $\mathbf{Pb} = \begin{bmatrix} 1/2 & 1/2 & 0 \\ 1/2 & 1/2 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ 3 \\ 4 \end{bmatrix}$

$$= \begin{bmatrix} 5/2 \\ 5/2 \\ 4 \end{bmatrix}$$